

Title

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This manuscript was compiled on August 27, 2026

Abstract

Significance

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Many quantitative genetic models built on the classical Darwinian paradigm assume a direct genotype-phenotype mapping, in which an individual's phenotype is determined by its own genotype and a generic, non-social environment (1). However, this assumption is challenged once social interactions are taken into account, because part of the environment is made up of the genotypes and phenotypes of interacting conspecifics (2, 3). When organisms interact with each other, certain phenotype may be affected depending on who the interacting partner is and what the nature of the interaction is. For example, . Such a phenotype is termed a social interacting phenotype, a trait whose expression is affected by interactions with conspecifics (2).

In order to understand the evolutionary consequences of such social interacting phenotype, Moore et al. proposed a quantitative genetic model that partitioned the phenotype into a direct genetic contribution from one's own genotype, and an indirect genetic contribution from a social partner genotype (2). They predicted that the evolutionary response of a social

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interacting phenotype should differ, often be greater, compared to a nonsocial phenotype, especially when the same trait is affected between social partners reciprocally, creating a feedback loop that drives phenotype evolution. Many extensions of this model have since been developed within the quantitative genetic framework (4–9) and simulation (10). Empirical studies have estimated the value of the proposed interaction coefficient, ψ , in empirical systems (11, 12). Despite its theoretical importance as a central parameter to indirect genetic effect models, it is more difficult to measure ψ than using other approaches such as variance partitioning, to measure the effect of social partners' genotype on one's phenotype (13).

Despite its influence, Moore et al.'s framework leaves two aspects of social interaction trait evolution underspecified. First, the model does not formalize who a focal individual interacts with or how that structure is determined; when extended to larger group size, many models assumed the average of all individuals in the group is influencing the focal individual (5). This gap can be filled with evolutionary social learning theory which provides a formal definition for who, when and what individuals copy, known as social learning strategies (14–16). Second, Moore et al.'s quantitative genetic framework does not specify an explicit genetic architecture linking genotype to the interacting phenotype. Such omission misses the connection to a large body of empirical studies which infer selection on social interacting phenotype from molecular signature like dN/dS (17–21). In order to make the comparison, explicit mapping between genotype and phenotype is required in the modeling to generate falsifiable predictions about the evolutionary path of an interacting trait.

In this study, we built a population genetic individual-based model with explicit genetic architecture to test how social learning strategies – copy-the-average, copy-the-best, or copy-the-extreme – affects the evolution of a socially interacting trait. For each strategy, we varied the strength of social interaction and compared social and asocial populations under three selection regimes: stabilizing, directional, and fluctuating selection. In addition, we used an invasion analysis to test whether each strategy can invade an asocial resident population, a prerequisite for a fully social population to arise in the first place. We found that whether sociality helps or hinders adaptation depends on the interaction strategy, strength of interaction, and the stability of the environment.

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